

DEPARTMENT OF MANAGEMENT
ETHIOPIAN HIGHER EDUCATION EXIT EXAM
(BA DEGREE IN MANAGEMENT)

Course Title: Operations Management

Time Allowed: 1:00 Hrs

Maximum Mark: 100%

Dear Prospective Graduates,

We have designed the following questions to assist you in preparing for your Ethiopian Higher Education Exit Exam. Please note that answering these questions does not guarantee a passing score on your exam. We strongly advise you to put in the necessary effort and dedication to achieve success.

Best of luck in your studies!

1. Which of the following best defines operations management?
 - A. The process of managing the marketing and sales activities of a company
 - B. The discipline that focuses on the design, planning, and control of production and operational processes
 - C. The management of human resources within an organization
 - D. The financial management of a company's operations and investments
2. What is the primary goal of operations management?
 - A. Maximizing shareholder wealth
 - B. Minimizing costs
 - C. Maximizing customer satisfaction
 - D. Optimizing employee performance
3. Which of the following is NOT within the scope of operations management?
 - A. Designing production processes
 - B. Managing the supply chain
 - C. Developing marketing strategies
 - D. Improving product quality
4. Which of the following activities is associated with capacity planning in operations management?
 - A. Setting quality standards for products
 - B. Determining the optimal level of resources required to meet customer demand
 - C. Monitoring employee performance on the shop floor
 - D. Analyzing market trends and customer preferences
5. Which term refers to the process of identifying and eliminating waste in operations management?
 - A. Process improvement
 - B. Inventory management
 - C. Quality control
 - D. Capacity planning

6. What is the purpose of inventory management in operations management?
 - A. To ensure efficient production processes
 - B. To minimize production costs
 - C. To optimize customer satisfaction
 - D. To manage supplier relationships
7. Which of the following best describes the role of operations management in an organization?
 - A. Managing financial resources
 - B. Leading marketing campaigns
 - C. Overseeing production processes and resource allocation
 - D. Developing strategic business plans
8. Which methodology is commonly used in operations management to identify and eliminate inefficiencies?
 - A. Total Quality Management (TQM)
 - B. Cost-Benefit Analysis (CBA)
 - C. Supply Chain Management (SCM)
 - D. Just-In-Time (JIT)
9. What is the primary focus of supply chain management in operations management?
 - A. Maximizing production capacity
 - B. Minimizing product defects
 - C. Optimizing transportation logistics
 - D. Increasing customer loyalty
10. Which of the following is an essential characteristic of effective operations management?

A. Reactive decision-making	C. Continuous improvement
B. Insufficient resource allocation	D. Limited employee involvement
11. Operations strategy refers to:
 - A. The overall direction and goals of the operations function within an organization
 - B. The marketing and sales strategies implemented by an organization
 - C. The financial strategies employed by an organization
 - D. The human resource management strategies within an organization
12. Competitiveness in operations management can be defined as:
 - A. The ability to produce goods or services at the lowest cost possible
 - B. The ability to outperform competitors in terms of quality, delivery, and innovation
 - C. The ability to maintain a high level of customer satisfaction
 - D. The ability to maximize profitability and shareholder value
13. Which of the following factors is NOT typically considered when formulating an operations strategy?
 - A. Market demand and customer expectations
 - B. Competitor analysis and benchmarking
 - C. Financial performance and budgetary constraints
 - D. Human resource management policies

14. The decision-making process involved in designing the operations system includes:
- A. Capacity planning, inventory management, and supply chain optimization
 - B. Pricing strategies, advertising campaigns, and market research
 - C. Hiring and training employees, performance evaluation, and compensation management
 - D. Financial forecasting, budget allocation, and risk management
15. Capacity planning involves:
- A. Determining the optimal level of production resources required to meet customer demand
 - B. Managing the flow of materials and information throughout the supply chain
 - C. Improving the efficiency and effectiveness of production processes
 - D. Analyzing market trends and customer preferences
16. Which of the following decisions is part of controlling the operations system?
- A. Selecting suppliers and managing vendor relationships
 - B. Setting quality standards and implementing quality control measures
 - C. Designing the layout and flow of the production facility
 - D. Developing marketing strategies and promotional campaigns
17. Which of the following is an example of an operations strategy aimed at achieving cost leadership?
- A. Implementing a continuous improvement program to reduce waste and increase efficiency
 - B. Investing in research and development to create innovative products
 - C. Differentiating products through superior quality and customization options
 - D. Expanding market reach through aggressive advertising and pricing strategies
18. Just-in-Time (JIT) is an operations strategy that focuses on:
- A. Minimizing inventory levels and carrying costs by receiving materials only when needed
 - B. Maximizing production capacity by maintaining large inventories of raw materials
 - C. Implementing a flexible production system that can quickly adapt to changes in customer demand
 - D. Outsourcing production activities to third-party suppliers to reduce costs
19. Quality control measures in operations management aim to:
- A. Minimize defects and ensure that products meet or exceed customer expectations
 - B. Maximize production output by reducing setup times and cycle times
 - C. Optimize transportation logistics and improve supply chain efficiency
 - D. Develop effective marketing campaigns and promotional strategies
20. Which of the following decisions is involved in supply chain management?
- A. Selecting the most cost-effective production technology and equipment
 - B. Determining the optimal order quantity and reorder point for inventory management
 - C. Negotiating contracts with suppliers and managing supplier relationships
 - D. Designing the layout and flow of the production facility

21. In operations management, the term "lean manufacturing" refers to:
- A. Minimizing waste and maximizing efficiency through the elimination of non-value-added activities
 - B. Implementing a just-in-time inventory system to reduce carrying costs
 - C. Increasing product variety and customization options to meet diverse customer needs
 - D. Maximizing output by maintaining high levels of inventory and production capacity
22. The decision to automate production processes is an example of:
- A. Capacity planning
 - B. Supply chain optimization
 - C. Quality control
 - D. Process improvement
23. Facility location decisions involve determining:
- A. The optimal arrangement of resources within a facility
 - B. The best layout for a facility
 - C. The most suitable geographic location for a facility
 - D. The specific equipment and machinery needed in a facility
24. Which of the following factors is NOT typically considered when making facility location decisions?
- A. Proximity to customers and suppliers
 - B. Availability of skilled labor
 - C. Transportation costs
 - D. Production capacity requirements
25. A company wants to open a new distribution center to serve its customers. Which location selection method would be most suitable for this decision?
- A. Factor rating method
 - B. Center-of-gravity method
 - C. Break-even analysis
 - D. Pareto analysis
26. In facility layout planning, the objective is to:
- A. Maximize production capacity
 - B. Minimize inventory holding costs
 - C. Minimize transportation costs
 - D. Optimize the flow of materials and information
27. Which of the following layout types is best suited for high-volume, standardized production?
- A. Process layout
 - B. Product layout
 - C. Cellular layout
 - D. Fixed-position layout
28. The objective of aggregate planning is to:
- A. Determine the optimal facility location
 - B. Optimize the layout of a facility
 - C. Allocate resources to meet demand over a specified time horizon
 - D. Minimize transportation costs in the supply chain
29. Which of the following quantitative techniques can be used for aggregate planning?
- A. Linear programming
 - B. Quality control charts
 - C. Pareto analysis
 - D. Break-even analysis

30. A company is using linear programming for aggregate planning. The decision variables represent the production quantities for different products. What type of problem is this?
- A. Integer programming
 - B. Continuous programming
 - C. Binary programming
 - D. Network programming
31. Which of the following is NOT a key consideration in aggregate planning?
- A. Demand forecast
 - B. Production capacity
 - C. Lead times
 - D. Material handling costs
32. The transportation method is commonly used in aggregate planning to:
- A. Determine the optimal facility location
 - B. Optimize the layout of a facility
 - C. Allocate resources to meet demand at minimum cost
 - D. Calculate the breakeven point in production
33. Which of the following is a benefit of using simulation in facility layout planning?
- A. Quick and simple analysis of layout alternatives
 - B. Accurate prediction of future demand
 - C. Ability to consider complex interactions between layout elements
 - D. Cost-effective solution for small-scale facilities
34. Which of the following is a tool used for analyzing the efficiency of a facility layout?
- A. Fishbone diagram
 - B. Value stream mapping
 - C. Kanban system
 - D. Gantt chart
35. Which of the following methods is used to determine the optimal order quantity in inventory management?
- A. Economic Order Quantity (EOQ)
 - B. Just-in-Time (JIT)
 - C. Material Requirements Planning (MRP)
 - D. Kanban
36. Which of the following factors is NOT considered a dimension of service quality in operations management?
- A. Reliability
 - B. Responsiveness
 - C. Cost-effectiveness
 - D. Empathy
37. Which of the following production strategies focuses on producing a wide variety of customized products with low volume and high flexibility?
- A. Mass production
 - B. Job production
 - C. Batch production
 - D. Continuous production
38. Which of the following is NOT a key principle of lean manufacturing in operations management?
- A. Just-in-Time (JIT) production
 - B. Total Quality Management (TQM)
 - C. Continuous improvement
 - D. Elimination of waste
39. Which of the following forecasting methods is based on the assumption that historical patterns will continue into the future?
- A. Moving average
 - B. Exponential smoothing
 - C. Delphi method
 - D. Regression analysis

40. Which of the following inventory management techniques focuses on classifying inventory items based on their importance and allocating resources accordingly?
- A. ABC analysis
 - B. Economic Order Quantity (EOQ)
 - C. Material Requirements Planning (MRP)
 - D. Kanban system
41. Which of the following performance metrics is used to measure the overall efficiency and effectiveness of a manufacturing process?
- A. Cycle time
 - B. Throughput
 - C. Utilization
 - D. Overall Equipment Efficiency (OEE)
42. A manufacturing company has a production line with a total capacity of 1,000 units per day. The average demand for the company's product is 800 units per day. What is the utilization rate of the production line?
- A. 80%
 - B. 100%
 - C. 120%
 - D. 125%
43. A company wants to improve its production process and reduce defects. The current defect rate is 2%. After implementing a quality improvement program, the defect rate drops to 1%. What is the percentage reduction in defects?
- A. 50%
 - B. 67%
 - C. 75%
 - D. 100%
44. A company wants to determine the total cost of producing a batch of 1,000 units. The fixed costs are \$10,000, and the variable costs per unit are \$5. What is the average cost per unit for this batch?
- A. \$10
 - B. \$15
 - C. \$20
 - D. \$25
45. In a production process, the cycle time is 2 minutes, and the work-in-progress (WIP) inventory is 100 units. What is the throughput rate of the process?
- A. 50 units per minute
 - B. 100 units per minute
 - C. 200 units per minute
 - D. 500 units per minute
46. A company produces 1,000 units of a product per day with an average defect rate of 3%. If the company wants to achieve a defect rate of 1%, how many defects per day should be eliminated?
- A. 10 defects
 - B. 20 defects
 - C. 30 defects
 - D. 40 defects
47. A manufacturing process has a capacity of 500 units per hour. The process operates for 10 hours per day, but due to maintenance and other downtime, the effective operating time is 80% of the total available time. What is the daily production capacity of this process?
- A. 4,000 units
 - B. 4,500 units
 - C. 5,000 units
 - D. 5,500 units

48. A company is considering two alternative production processes. Process A has fixed costs of \$50,000 and variable costs of \$10 per unit. Process B has fixed costs of \$30,000 and variable costs of \$15 per unit. If the company expects to produce 5,000 units, which process should be chosen based on cost considerations?
- A. Process A
 - B. Process B
 - C. Both processes have the same cost
 - D. More information is needed to determine the cost
49. A company has a lead time of 7 days to receive raw materials. The company wants to maintain a safety stock that will cover the demand during the lead time with a 95% service level. If the average daily demand is 100 units and the standard deviation of demand is 20 units, what is the safety stock quantity?
- A. 100 units
 - B. 140 units
 - C. 180 units
 - D. 220 units
50. A company's production line has a total downtime of 5 hours during a 24-hour period. If the line operates at a 90% efficiency rate, what is the total available production time?
- A. 18 hours
 - B. 19 hours
 - C. 20 hours
 - D. 21 hours

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OPERATIONS MANAGEMENT
MODEL EXIT EXAM ANSWER SHEET
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